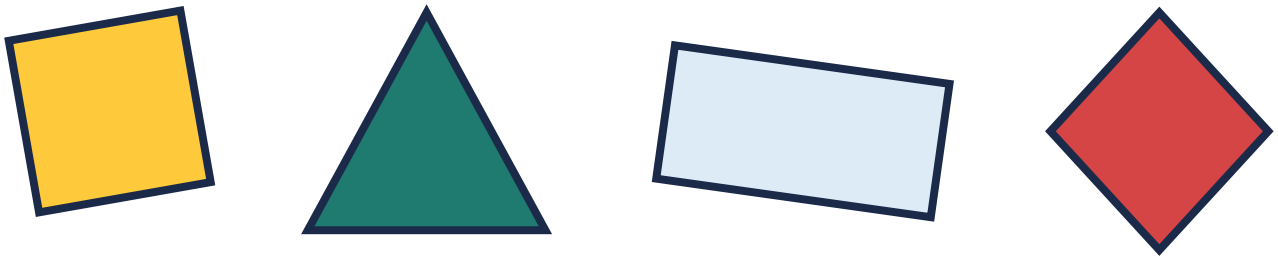
FOR PARENTS • PRIMARY 2 HOLIDAY PROGRAM

# Weekly Dose of Thinking Math

One dose a week for five weeks: short sets of questions for every Math Inquirers lesson, a tip for you, and the answers at the back.



**Notice it. Test it. Explain why.**

Math Inquirers Holiday Program
Preciouslyours Pte Ltd

BEFORE YOU START

# 00 FOR GROWN-UPS How to help

1. **Ask before you tell.** Start with "What do you notice?" and let your child answer first.
2. **Use real things.** Coins, notes, cups, string, paper strips or blocks. Try it with objects or a drawing before writing.
3. **Give hints in three steps.** Look again. Try it with objects. Try a smaller example. Say the answer last.
4. **Ask "How do you know?"** even when the answer is right. The reason matters as much as the answer.
5. **Praise the thinking.** "I like how you checked that" works better than "You are clever".
6. **Keep it short.** Ten minutes and two or three questions is enough. Stop while it is still fun.
7. **Answers are at the back.** Look only after your child has tried.

> **Hard question? Great. Getting stuck is where the thinking happens.**

## One dose a week

Each week has two lessons and about 18 questions. Do two or three a day, Monday to Friday, and keep the weekend for the Challenge questions if your child enjoys them.

## The three levels

**Core** is for everyone. **Think Harder** stretches the idea. **Challenge** is for strong thinkers, and it is fine to leave it for another day.

## What your child is doing in class

Ten 90-minute live online lessons over five weeks, in small groups. Each lesson starts with a hands-on puzzle, then a short teaching part, then the workbook. These pages practise the same ideas at home.

# W1 Digit Detectives

LESSON 1 • NUMBERS TO 1000

## Core

1 What number is 5 hundreds, 3 tens and 8 ones?

2 In 914, what is the digit 1 worth?

3 What number is 100 more than 457?

4 What number is 10 less than 603?

## Think Harder

5 Use the digit cards 1, 5 and 8 once each. How many different 3-digit numbers can you make? List them in order.

6 Use the digit cards 4, 4 and 7. How many different 3-digit numbers can you make? Why is it not 6?

7 What number is 3 hundreds, 14 tens and 6 ones?

### What your child learns

Building 3-digit numbers from hundreds, tens and ones, finding every number from digit cards by listing in order, and spotting patterns on a 1000 chart.

### Tip for you

Write three digits on paper scraps. Ask your child to make the biggest and smallest numbers, then every number, in order. Before class: cut out the digit cards (kit page) and print or draw a 1000 chart with rows of 10.

### Ask

"How do you know you have found them all?"

## Challenge

**8** Use any three of the digit cards 1, 3, 5 and 7. How many different 3-digit numbers can you make?

**9** How many 3-digit numbers have digits that add up to 5?

# W1 Missing Digit Mysteries

LESSON 2 • ADDITION AND SUBTRACTION WITHIN 1000

## Core

1 3A5 + 214 = 579. What is the digit A?

2 6A8 − 205 = 433. What is the digit A?

3 What number makes this true? 400 + 250 = 300 + A

4 What must be added to 365 to make 500?

## Think Harder

5 Find the missing digits: 5A7 + B48 = 905

6 Find the missing digits: 7A1 − B58 = 263

7 What number makes this true? 538 + 199 = 540 + A

### What your child learns

Finding missing digits in sums and differences, checking by adding back, and magic squares with bigger numbers.

### Tip for you

Cover one digit of a sum with your thumb. Let your child work out what is hiding, then check by adding. Before class: have paper scraps and a pencil ready for the number tiles 110 to 160.

### Ask

"How can you check your answer?"

## Challenge

**8** Use the digit cards 1, 2, 3, 4, 5 and 6 once each to make two 3-digit numbers. Make the difference as BIG as possible.

**9** The same number A goes in both places: A + 150 = 550 – A. What is A?

# W2 Legs, Wheels and Arrays

LESSON 3 • MULTIPLICATION AND DIVISION

## Core

1 How many wheels are there on 5 bicycles?

2 There are 4 rows of chairs with 5 chairs in each row. How many chairs are there?

3 18 sweets are shared equally among 3 children. How many sweets does each child get?

4 How many legs do 7 hens have?

## Think Harder

5 There are 9 hens and goats on a farm. They have 24 legs altogether. How many hens and how many goats are there?

6 Ali shares 34 marbles equally among 4 friends, giving as many as he can. How many does each friend get? How many are left over?

### What your child learns

Arrays, equal groups and sharing with leftovers, using the 2, 3, 4, 5 and 10 times tables.

### Tip for you

Count legs at the dinner table or wheels in the car park. Ask how many people or cars there must be. Before class: gather about 30 counters (buttons, dried beans or Lego bricks) and paper.

### Ask

"What do you do with the ones left over?"

**7** An array of stickers has 4 rows. Ben counts 12 stickers in the first 3 rows. How many stickers are in the whole array?

## Challenge

**8** Ali says: “Every number from 4 to 30 can be made into an array with at least 2 rows and at least 2 counters in each row.” Is he right? Find ALL the numbers that cannot. Explain why they cannot.

**9** A shop sells bicycles (2 wheels), tricycles (3 wheels) and toy cars (4 wheels). There are 9 vehicles and 28 wheels altogether, with at least one of each kind. Find ALL the ways.

# W2 Pattern Machines

LESSON 4 • PATTERNS WITH MULTIPLICATION

## Core

1 Count in 4s. Fill in the next three numbers: 4, 8, 12, __, __, __

2 A pattern goes red, blue, red, blue ... What colour is the 10th bead?

3 Each hand has 5 fingers. How many fingers are on 6 hands?

4 A pattern goes star, circle, circle, star, circle, circle ... What is the 9th shape?

## Think Harder

5 A pattern goes A, B, B, C, A, B, B, C ... What is the 25th letter?

6 Beads go red, yellow, yellow, red, yellow, yellow ... How many yellow beads are in the first 18 beads?

### What your child learns

Finding far-away places in a repeating pattern by counting in groups, and growing patterns with a table.

### Tip for you

Make a pattern with spoons and forks. Ask what the 20th one will be before you build it. Before class: gather 9 red and blue cubes (or Lego bricks), about 25 lolly sticks and coloured pencils.

### Ask

"How many things are in the part that repeats?"

**7** Houses are made of sticks in a row. 1 house uses 5 sticks, 2 houses use 9 sticks, 3 houses use 13 sticks. How many sticks are needed for 6 houses?

## Challenge

**8** Light A flashes every 3 seconds. Light B flashes every 5 seconds. They flash together at the start. In the next 30 seconds, when do they flash together?

**9** Houses are made of sticks in a row: 1 house uses 5 sticks, 2 houses use 9, 3 houses use 13. Ben has 40 sticks. What is the most houses he can make? How many sticks are left over?

# W3 LESSON 5 • WORKING BACKWARDS
# Mystery Number Machines

## Core

**1** I think of a number. I multiply it by 5. I get 45. What is my number?

**2** I think of a number. I divide it by 2. I get 8. What is my number?

**3** I think of a number. I add 35. I get 90. What is my number?

**4** I think of a number. I multiply it by 3, then add 2. I get 20. What is my number?

## Think Harder

**5** A machine does × 4, then + 13. The OUT is 45. What went IN?

**6** Ben had some money. He spent $15 on a book. Then his dad gave him $10. Now he has $42. How much money did Ben have at first?

### What your child learns

Two-step number machines and working backwards from the answer, with bar models.

### Tip for you

Play "I double my number and add 3, and get 15". Swap roles and let your child make up the machine. Before class: have paper strips and counters ready, and write x cards (x 2, x 3, x 4, x 5, x 10) and + or - cards (1 to 20) on small pieces of paper.

### Ask

"What is the opposite of each step?"

**7** Siti had some beads. She gave half of them to Mei. Then she bought 12 more. Now she has 30 beads. How many beads did Siti have at first?

## Challenge

**8** Ali, Ben and Siti share 42 stickers. Ben gets 2 more than Ali. Siti gets twice as many as Ali. How many stickers does each child get?

**9** A machine does × 2, then − 4. Ravi puts a number in, then puts the OUT straight back in again. The final OUT is 12. What number did he put in first?

# W3 LESSON 6 • MONEY Market Day

## Core

**1** Write 275¢ in dollars and cents.

**2** Pay exactly 65¢ with the fewest coins.

**3** Pay exactly $17 with the fewest notes.

**4** A pencil costs 45¢. Mei pays with a $1 coin. How much change does she get?

## Think Harder

**5** How many ways can you pay exactly $10 using $5 notes, $2 notes and $1 coins?

**6** Ali has 4 coins. They are worth $1.25 altogether. Find all the sets of 4 coins he could have.

**7** How many different ways can you pay exactly 40¢ using 5¢, 10¢ and 20¢ coins? Make a list in order, starting with the most 20¢ coins.

### What your child learns

Paying with the fewest notes and coins, finding every way to pay, giving change and checking a budget.

### Tip for you

Let your child pay at a hawker stall or shop and count the change before you check it. Before class: cut out the play coins and notes (kit pages) and keep the spare coins in the envelope.

### Ask

"Is there a way with fewer coins?"

## Challenge

**8** Ali has 10 coins. Every coin is a 10¢ coin or a 20¢ coin. They could be worth $1.50: how many of each coin is that? Write ALL the totals his 10 coins could be worth. Could they be worth $1.55? Explain.

**9** Mei has $1. At the snack stall: chips 45¢, sweet 15¢, jelly 25¢, biscuit 30¢, lollipop 20¢. She wants as many DIFFERENT snacks as she can. Which should she buy?

# W4 LESSON 7 • FRACTIONS
# Fair Shares

## Core

**1** A cake is cut into 5 equal pieces. Ben eats 2 pieces. What fraction of the cake does he eat?

**2** How many quarters make 1 whole?

**3** Which is bigger, 1/3 or 1/6?

**4** 3/10 + 4/10 = ?

## Think Harder

**5** Write these in order, smallest first: 1/3, 1/12, 1/2, 1/5.

**6** A ribbon is cut into 9 equal pieces. Mei uses 4 pieces and Siti uses 3 pieces. What fraction of the ribbon is left?

**7** 2/7 + ? = 1 whole

### What your child learns

Halves, thirds and quarters as equal parts, comparing fractions, and adding parts of the same whole.

### Tip for you

Share a roti prata or a pizza fairly. Ask if the pieces are really equal, and which is bigger: a third or a quarter. Before class: cut out the fraction strips (kit page) and have 4 paper strips of the same length.

### Ask

"Are the parts the same size?"

## Challenge

**8** Three children share a cake cut into 10 equal pieces. Each child gets a DIFFERENT number of pieces (at least 1), and the whole cake is shared out. Find ALL the ways to share it. Write each child's share as a fraction.

**9** I am a fraction with 1 on top. I am smaller than 1/4 but bigger than 1/7. What could I be? Find all the answers.

# W4 LESSON 8 • 2D AND 3D SHAPES
# Shape Builders

## Core

**1** How many quarter circles make a whole circle?

**2** How many semicircles make a whole circle?

**3** How many faces does a cube have?

**4** How many squares of ALL sizes are in a grid 2 squares long and 2 squares high? (Small ones and one big one.)

## Think Harder

**5** A stack of cubes has a bottom layer 4 cubes long and 3 cubes deep, a middle layer 3 cubes long and 2 cubes deep, and 1 cube on top. How many cubes are in the stack? Draw the layers.

**6** How many squares are in a grid 3 squares long and 2 squares high?

**7** A stack has 2 layers. Each layer is 3 cubes long and 2 cubes wide. How many cubes?

### What your child learns

Making and spotting shapes, semicircles and quarter circles, counting cubes in stacks, and nets.

### Tip for you

Build a small tower with blocks or sugar cubes. Ask how many cubes there are, including the ones you cannot see. Before class: have scissors, 2 paper circles of the same size and about 12 small cubes ready, and cut out nets A to F (kit page).

### Ask

"Did you count the hidden ones?"

## Challenge

**8** How many squares can you find in a grid 4 squares long and 3 squares high? Another grid, longer than it is high and at least 2 squares high, also has exactly 20 squares of all sizes. Find ALL the grids like that.

**9** A square has both lines from corner to corner drawn. How many triangles can you find?

# W5 Measure Masters

LESSON 9 • LENGTH, MASS, VOLUME AND TIME

## Core

**1** One bag of flour balances 2 bags of sugar. A bag of sugar is 300 g. How heavy is the bag of flour?

**2** A bucket holds 10 ℓ. It has 4 ℓ of water in it. Ali pours in 3 ℓ more. How many more litres are needed to fill it?

**3** A ribbon is 90 cm long. Mei cuts off 25 cm. How long is the piece that is left?

**4** A cartoon starts at 5:40 p.m. and lasts 30 minutes. What time does it end?

## Think Harder

**5** 3 apples balance 1 pomelo. The pomelo and 2 apples together balance a 500 g weight. How heavy is one apple? How heavy is the pomelo?

**6** You have a 2 ℓ jug and a 7 ℓ jug with no marks. How can you get exactly 3 ℓ? List each pour.

### What your child learns

Balance scales, pouring puzzles with jugs, lengths end to end, and telling time and how long things take.

### Tip for you

Use two cups of different sizes and a basin. Can your child measure exactly one cup of water a different way? Before class: make the coat-hanger balance (hanger, two paper cups, string), have a glue stick, a spoon and 20 coins, two different-size cups with a jug of dry rice or sand (tape marks at 3 and 5), and cut out the paper clock.

### Ask

"What do you know for sure from the scale?"

**7** Mei's piano lesson ends at 4:15 p.m. It lasted 45 minutes. What time did it start?

## Challenge

**8** A basket with 4 apples is 700 g. The same basket with 2 apples is 500 g. How heavy is the empty basket?

**9** A rope 40 cm long is cut into pieces 10 cm long. Each cut takes 2 minutes. How long does the cutting take?

# W5 Challenge Day

LESSON 10 • MIXED PUZZLES

## Core

**1** Use the digit cards 5, 1 and 9 once each. Make the greatest 3-digit number and the smallest 3-digit number.

**2** In a picture graph, each picture stands for 2 children. Apples has 4 pictures. How many children chose apples?

**3** Ali has 3 bags with 5 marbles in each bag. Mei has 4 bags with 4 marbles in each bag. Who has more marbles? How many more?

**4** Make 80¢ with exactly 3 coins. You can use 5¢, 10¢, 20¢ and 50¢ coins.

## Think Harder

**5** 30 children chose a fruit. In the picture graph each picture stands for 5 children. Apple has 2 pictures and banana has 3 pictures. The mango row is missing. How many pictures should the mango row have?

**6** How many 3-digit numbers have digits that add up to 3? List them all.

### What your child learns

Choosing how to start a puzzle: draw it, list it or make a table, plus picture graphs and logic grids.

### Tip for you

When your child is stuck, ask "Can you draw it?" before giving any other hint. Before class: have the digit cards, play coins, paper strips, a pencil and a small whiteboard ready.

### Ask

"How did you start? What would you try next time?"

**7** A bus had some people on it. At the first stop half of the people got off. At the second stop 6 people got on. Now there are 14 people on the bus. How many people were on the bus at the start? Draw a bar to start.

## Challenge

**8** Use three DIFFERENT digit cards to make 3-digit numbers. With the cards 0, 3 and 8 you can make exactly 4 numbers. With the cards 3, 5 and 8 you can make 6. Which sets of three different cards make exactly 4 numbers? Test some sets and write a rule. Does your rule always work?

**9** Ali, Mei, Siti and Ben each go to school a different way: bus, MRT, car or walking. Clues: Siti walks. Ben does not go by bus or by car. Mei does not go by car. Who goes by what? Then take away ONE clue at a time. Which clue could you take away and still have exactly one answer? How many answers does each clue-less puzzle have?

LOOK AFTER YOUR CHILD HAS TRIED

ALL LESSONS

# Answers

## Week 1 • Lesson 1: Digit Detectives

1. 538
2. 10 (1 ten)
3. 557
4. 593
5. 6 numbers: 158, 185, 518, 581, 815, 851.
6. 3 numbers: 447, 474, 744. The two 4s look the same, so swapping them does not make a new number.
7. 446 (14 tens = 1 hundred and 4 tens).
8. 24. Leave out one card (4 choices); each set of 3 cards makes 6 numbers: 6 + 6 + 6 + 6 = 24.
9. 15. Hundreds digit 1: 5 numbers; 2: 4; 3: 3; 4: 2; 5: 1. 5 + 4 + 3 + 2 + 1 = 15.

## Week 1 • Lesson 2: Missing Digit Mysteries

1. 6 (365 + 214 = 579)
2. 3 (638 – 205 = 433)
3. 350
4. 135
5. 557 + 348 = 905 (ones 7 + 8 = 15; tens 5 + 4 + 1 = 10; hundreds 5 + 3 + 1 = 9).
6. 721 – 458 = 263. Check: 263 + 458 = 721.
7. 197. 538 is 2 less than 540, so A is 2 less than 199. Both sides are 737.
8. 531: 654 – 123.
9. 200 (200 + 150 = 350 and 550 – 200 = 350).

## Week 2 • Lesson 3: Legs, Wheels and Arrays

1. 10 wheels (2, 4, 6, 8, 10).
2. 20 chairs (4 × 5 = 20).
3. 6 sweets (18 ÷ 3 = 6).
4. 14 legs (7 × 2 = 14).
5. 6 hens and 3 goats. Start with 9 hens (18 legs); each swap to a goat adds 2 legs; 3 swaps make 24. Check: 12 + 12 = 24.
6. 8 each (4 × 8 = 32) and 2 left over.
7. 16 stickers. 3 rows have 12, so each row has 4 (12 ÷ 3 = 4); 4 rows of 4 is 16.
8. No. 5, 7, 11, 13, 17, 19, 23 and 29 cannot (8 numbers): only 1 row works.
9. 3 ways (bikes, trikes, cars): 3, 2, 4; 2, 4, 3; 1, 6, 2.

## Week 2 • Lesson 4: Pattern Machines

1. 16, 20, 24
2. Blue (every even place is blue).
3. 30 fingers (5, 10, 15, 20, 25, 30).
4. A circle. 9 ends a group of 3, and each group ends with a circle.
5. A. 24 ends a group of 4 (4 × 6 = 24), so 25 starts again with A.
6. 12 yellow beads (18 beads is 6 groups of 3, with 2 yellow in each).
7. 25 sticks. The rule is add 4: 5, 9, 13, 17, 21, 25.
8. At 15 seconds and at 30 seconds (numbers in both the 3 and 5 times tables).
9. 9 houses use 37 sticks, with 3 left over (10 houses would need 41).

## Week 3 • Lesson 5: Mystery Number Machines

1. 9
2. 16
3. 55
4. 6 (20 – 2 = 18, 18 ÷ 3 = 6)
5. 8 (45 – 13 = 32, 32 ÷ 4 = 8)
6. $47 (42 – 10 = 32, 32 + 15 = 47)
7. 36 (30 – 12 = 18 is half, 18 + 18 = 36)
8. Ali 10, Ben 12, Siti 20 (40 ÷ 4 = 10)
9. 6 (back twice: 12 → 8 → 6)

## Week 3 • Lesson 6: Market Day

1. $2.75
2. 50¢ + 10¢ + 5¢ (3 coins)
3. $10 + $5 + $2 (3 notes)
4. 55¢
5. 10 ways (1 + 3 + 6)
6. 2 sets: $1 + 10¢ + 10¢ + 5¢, and 50¢ + 50¢ + 20¢ + 5¢
7. 9 ways: two 20¢ (1 way); one 20¢ and 20¢ more (3 ways); no 20¢ (5 ways).
8. Five and five. Totals $1.00 to $2.00, 10¢ apart (11). Not $1.55.
9. Sweet, lollipop, jelly and biscuit (90¢). Five snacks cost $1.35.

## Week 4 • Lesson 7: Fair Shares

1. 2/5
2. 4
3. 1/3 (3 parts are bigger than 6 parts)
4. 7/10
5. 1/12, 1/5, 1/3, 1/2
6. 2/9 (4 + 3 = 7 pieces used, 9 – 7 = 2 left)
7. 5/7
8. 4 ways (tenths): 1, 2, 7; 1, 3, 6; 1, 4, 5; 2, 3, 5.
9. 1/5 and 1/6

## Week 4 • Lesson 8: Shape Builders

1. 4
2. 2
3. 6
4. 5 (4 small + 1 big)
5. 19 (12 + 6 + 1)
6. 8 (6 small + 2 made of 4 small ones)
7. 12
8. 20 (12 + 6 + 2). Also a 7 by 2 grid.
9. 8 (4 small + 4 made of 2 small ones)

## Week 5 • Lesson 9: Measure Masters

1. 600 g (300 + 300)
2. 3 ℓ (4 + 3 = 7, and 10 – 7 = 3)
3. 65 cm
4. 6:10 p.m.
5. Apple 100 g, pomelo 300 g (5 apples balance 500 g).
6. Fill 7, pour into 2, empty 2, pour again: 3 ℓ is left in the 7 ℓ jug.
7. 3:30 p.m. (count back 15 minutes to 4:00, then 30 more minutes to 3:30)
8. 300 g (the 2 extra apples are 200 g)
9. 6 minutes (4 pieces need only 3 cuts)

## Week 5 • Lesson 10: Challenge Day

1. 951 and 159
2. 8 children
3. Mei, by 1 (Ali 15, Mei 16)
4. 50¢ + 20¢ + 10¢ (the only way)
5. 1 picture (30 – 10 – 15 = 5 children)
6. 6: 102, 111, 120, 201, 210, 300
7. 16 (14 – 6 = 8 is half, 8 + 8 = 16).
8. Exactly the sets with a 0 card: 0, a, b make a0b, ab0, b0a, ba0. Sets without 0 make 6.
9. Ali car, Mei bus, Siti walks, Ben MRT. No clue can go (8, 4 or 2 answers).